1. ________________command can be used by *workstation 'B'* to check *Server 'A'* existence?
A) ipconfig
B) ifconfig
C) tracert
D) ping

2. HTML is the Internet protocol used to transfer Web pages.
A) True
B) False

3. FTP is a protocol used to upload and download file to and from a server.
A) True
B) False

4. All of the following are basic technological foundations of the Internet except?
A) client/server computing
B) TCP/IP communications protocol
C) packet-switching hardware.
D) Circuit Switching

5. The process of slicing digital messages into parcels, sending them along different communication paths as they become available, and reassembling them at the destination point is called:
A) packet switching.
B) the Transmission Control Protocol.
C) routing.
D) the File Transfer Protocol.

6. Which of the following is the core communications protocol for the Internet?
A) OSI Protocol
B) HTTP
C) TCP/IP
D) FTP

7. The Transport Layer of TCP/IP is responsible for which of the following?
A) providing communication with the application by acknowledging and sequencing the packets to and from the application
B) placing packets on and receiving them from the network medium
C) providing a variety of applications with the ability to access the services of the lower layers
D) addressing, packaging, and routing messages

8. The Internet Layer of TCP/IP is responsible for ____________________?
A) providing communication with the application by acknowledging and sequencing the packets to and from the application
B) addressing, packaging, and routing messages
C) providing a variety of applications with the ability to access the services of the lower layers
D) placing packets on and receiving them from the network medium

9. The Network Interface Layer of TCP/IP is responsible for ______________?
A) providing communication with the application by acknowledging and sequencing the packets to and from the application
B) addressing, packaging, and routing messages
C) placing packets on and receiving them from the network medium
D) providing a variety of application with the ability to access the services of the lower layers

10. An IPv4 address is expressed as a:
A) 64-bit number that appears as a series of four separate numbers separated by semicolons.
B) 32-bit number that appears as a series of four separate numbers separated by periods.
C) 32-bit number that appears as a series of four separate numbers separated by semicolons.
D) 64-bit number that appears as a series of four separate numbers separated by periods.

11. ________________ is the natural language convention used to represent IP addresses:
A) Internet protocol addressing schema.
B) assigned numbers and names (ANN) system.
C) uniform resource locator system.
D) domain name system.

12. The addresses used by browsers to identify the location of content on the Web are called:
A) Hypertext Transfer Protocol.
B) Uniform resource locators.
C) IP addresses.
D) Domain names.

13. Which of the following statements about client/server computing is true?
A) It provided communication rules and regulations.
B) It connects multiple powerful personal computers together in one network with one or more servers dedicated to common functions that they all need.
C) It exploded the available communications capacity.
D) It does not provide sufficient computing capacity to support graphics or color.

14. Which of the following is *not* an advantage of client/server computing over centralized mainframe computing?
A) Each client added to the network increases the network's overall capacity and transmission speeds.
B) There is less risk that a system will completely malfunction because backup or mirror servers can pick up the slack if one server goes down.
C) It is easy to expand capacity by adding servers and clients.
D) Client/server networks are less vulnerable, in part because the processing load is balanced over many powerful smaller computers rather than concentrated in a single huge computer.

15. _______ is a model of computing in which firms and individuals obtain computing power and software applications over the Internet, rather than purchasing and installing it on their own computers.
A) Client/server computing
B) Broadband computing
C) P2P computing
D) Cloud computing

16. Which of the following protocols is used to send mail to a server?
A) Simple Mail Transport Protocol
B) Simple Mail Transfer Protocol
C) Simple Mail Transmission Protocol
D) Simple Mail Transaction Protocol

17. Where does FTP run within TCP/IP?
A) in the Internet Layer
B) in the Network Interface Layer
C) in the Transport Layer
D) in the Application Layer

18. ________ is one of the original Internet services and is used to transfer files from a server computer to a client computer and vice versa.
A) FTP
B) SMTP
C) SSL
D) HTTP

19. The ping command implements which protocol?
A) IGMP
B) IMCP
C) ICMP
D) IGRP

20. The set of technologies that enables efficient delivery of data to many locations on a network is called:
A) IP multicasting.
B) BigBand.
C) streaming.
D) diffserv.

21. ________ is a utility program that allows you to track the path of a message you send from your client to a remote computer on the Internet.
A) Tracert
B) Ping
C) Ipconfig
D) Telnet

22. A TCP/IP network located within a single organization for the purposes of communication and information processing is called a(n) ________.
A) LAN
B) Intranet
C) Internet
D) Private network

23. What is the Protocol Data Unit (PDU) employed at the Physical Layer?
   A) Bits
   B) Frames
   C) Packets
   D) segments

24. What layer of the OSI model is designed to perform error detection functions?
   A) Physical
   B) Data link
   C) Network
   D) Transport

25. What layer of the OSI model is associated with the MAC and LLC sublayers?
   A) Physical
   B) Data link
   C) Network
   D) Transport

26. Which Protocol Data Unit (PDU) is employed at the Transport Layer?
   A) Bits
   B) Frames
   C) Packets
   D) segments

27. Bits are packaged into frames at which layer of the OSI model?
   A) Physical
   B) Data link
   C) Network
   D) Transport

28. Which of the following is not a benefit of using a layered network model?
   A) it specifies how changes to one layer must be propagated through the other layers
   B) it facilitates troubleshooting
   C) it focuses on details rather than general functions of networking
   D) it breaks the complex process of networking into more manageable chunks
   E) it allows layers developed by different vendors to interoperate.

29. Which of the following does not operate at the presentation layer?
   A) Midi
   B) Mp3
   C) smtp
   D) jpeg

30. Flow control takes place at which layer?
   A) Physical
   B) Transport
   C) Network
   D) Mac sub-layer of the data link layer

31. Encryption takes place at which layer?
A) Application
B) Presentation
C) Session
D) Transport

32. The network layer uses physical addresses to route data to destination hosts.
A) True
B) False

33. Error detection and recovery takes place at which layer?
A) Transport
B) Presentation
C) Data link
D) Application

34. Which layer establishes, maintains, and terminates communications between applications located on different devices?
A) Transport
B) Presentation
C) Session
D) Application

35. IP is implemented at which OSI model layer?
A) Transport
B) Network
C) Session
D) Application

36. Which layer handles the formatting of application data so that it will be readable by the destination system?
A) Transport
B) Presentation
C) Session
D) Application

37. Which of the following is not presentation layer standards?
A) Midi
B) Tiff
C) Tftp
D) Gif

38. Packets are found at which layer?
    A) Transport
    B) Network
    C) Session
    D) Application

39. Which layer translates between physical and logical addresses?
    A) Network
    B) Data link
    C) Transport
    D) Session

40. What does OSI stand for?
    A) International Standards Organization
    B) Open Systems Interconnect
    C) Operating System Interconnection
    D) Open Systems Interface

41. Repeaters and hubs operate at which layer?
    A) Network
    B) Data link
    C) Transport
    D) Physical

42. Bit synchronization is handled at which layer?
    A) Network
    B) Data link
    C) Transport
    D) Physical

43. Most logical addresses are preset in network interface cards at the factory
    A) True
    B) False

44. Bridges operate at which layer of the OSI model?
    A) Network
    B) Data link
    C) Transport
    D) Physical

45. What are the sublayers of the data link layer?
   A) MAC and IPX
   B) Hardware and frames
   C) MAC and LLC
   D) WAN and LAN

46. Which layer translates between physical and logical addresses?
   A) Network
   B) Data link
   C) Transport
   D) Physical

47. Which layer is responsible for packet sequencing, acknowledgments, and requests for retransmission?
   A) Network
   B) Data link
   C) Transport
   D) Physical

48. The part of OSI where one most commonly finds data encryption, compression, and other encoding for network communication is -
   A) application
   B) session
   C) presentation
   D) none of above

49. Which of these network devices belongs to the OSI layer one?
   A) repeater
   B) router
   C) switch
   D) bridge

50. Which of these network devices belong to the OSI data link layer?
   A) repeater
   B) router
   C) switch
   D) gateway

<u>Use figure 1 below to answer questions 51 – 60.</u>

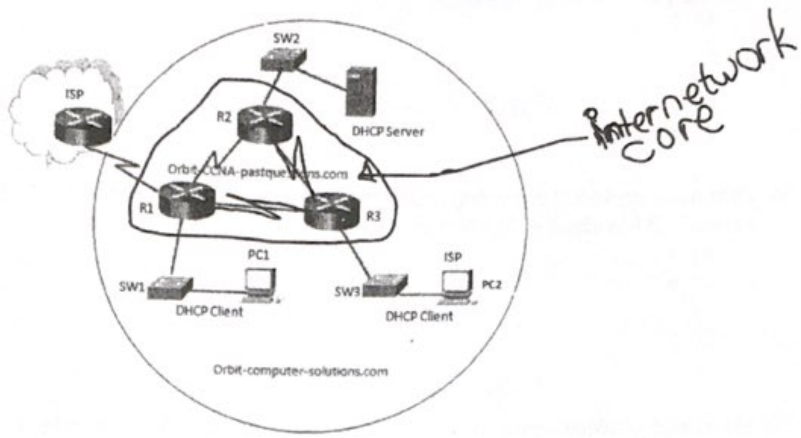

Figure 1: Network diagram

51. All hosts in the networks shown by figure 1 have been operational for several hours before the DHCP server went down. What happens to the hosts that have already obtained service from the DHCP server?
    - A) The hosts will not be able to communicate with any other hosts
    - B) The hosts will continue to communicate normally for a period of time
    - C) The hosts will be able to communicate with hosts outsides their own network
    - D) The hosts will only be able to communicate with other hosts by IP address not by hostname

52. How many broadcast domain(s) are in the network diagram shown by figure 1
    - A) 4
    - B) 6
    - C) 3
    - D) 7
    - E) 1

53. How many collision domain(s) are in the network diagram shown by figure 1
    - A) 6
    - B) 4
    - C) 3
    - D) 7
    - E) 5

54. How many collision domain(s) will be created by figure1 if SW1 is replaced with a hub and another two PC's named PC3, and PC4 are respectively patched to this hub
    - A) 4
    - B) 6
    - C) 3
    - D) 7
    - E) 5

55. How many collision domain(s) will be created by figure1 if SW1 is replaced with a bridge and another PC named PC3 is patched to this bridge

A) 4
B) 6
C) 3
D) 7
E) 5

56. How many broadcast domain(s) will be created by figure1 if SW1 is replaced with a bridge and another PC named PC3 is patched to this bridge

A) 4
B) 6
C) 3
D) 7
E) 5

57. How many collision domain(s) will be created by figure1 if SW1 is replaced with a hub and another switch named SW4 is patched to SW3

A) 4
B) 6
C) 3
D) 7
E) 5

58. What is the purpose of the DHCP server?

A) to provide storage for email
B) to translate URLs to IP addresses
C) to translate IPv4 addresses to MAC addresses
D) to provide an IP configuration information to hosts

59. How is the message sent from PC2 attempts to contact the DHCP Server when it is first powers on?

A) Layer 3 unicast
B) Layer 3 broadcast
C) Layer 3 multicast
D) Without any Layer 3 encapsulation
E) Layer 2 broadcast

60. What is the default behavior of router R1 when PC1 requests service (IP Address) from the DHCP server?

A) Drop the request
B) Forward the request to the DHCP server via a Broadcast message
C) Forward the request to the ISP to be forwarded to the DHCP server
D) Broadcast the request to all devices including the ISP

End of Exam

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